

2025 AMC 10B Problem 8 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 8

Problem:

Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was $\$ABB.BA$, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?

Answer Choices:

- A. 7
- B. 8
- C. 11
- D. 14
- E. 15

Join the Discussion

Stuck on this problem or want to share your approach?

Continue the conversation and see what others are thinking: [View Forum Thread](#) 

Solution:

The total cost is $\$ABB.BA$, with digits A and B , $A \neq 0$. In cents, let T represent the total cost of the sweatshirts which is the 5-digit integer $\overline{ABBA A}$.

Since there are 36 sweatshirts, the total in cents must be divisible by 36. Therefore T is divisible by 4 and by 9. Now, we can solve it in 2 ways, one of which uses modular arithmetic:

Solution 1: (Modular Arithmetic)

We know that the total price is divisible by 4 and 9. The last two digits are BA , so $10B + A$ is divisible by 4, so

$$10B + A \equiv 0 \pmod{4} \implies 2B + A \equiv 0 \pmod{4}. \quad (1)$$

Since the price is also divisible by 9, the sum of the digits is

$$A + B + B + B + A = 2A + 3B \equiv 0 \pmod{9} \quad (2)$$

From (2),

$$2A \equiv -3B \pmod{9} \implies A \equiv 5(-3B) \equiv -15B \equiv 3B \pmod{9}.$$

Testing $B = 0, 1, \dots, 9$ with $A \equiv 3B \pmod{9}$ and A a nonzero digit, the only pair (A, B) that also satisfies (1) is $(A, B) = (6, 5)$. Thus, the total cost is $\$655.56$, and

$$A + B = 6 + 5 = \boxed{\text{(C) } 11}$$

Solution 2: (Divisibility Rules):

The divisibility rule for 9 says that the sum of digits must be divisible by 9, so for some integer p

$$A + B + B + B + A = 2A + 3B = 9p$$

As $3B$ and $9p$ are divisible by 3, we know that A must also be divisible by 3. Furthermore, as $\overline{ABBB A}$ is divisible by 4, it is even, and hence A is even as well.

Since A is even, divisible by 3, and is a nonzero digit, $A = 6$. To find B , we note that since $2A + 3B = 9p$, our possibilities for B are 2, 5, 8. However, since $\overline{ABBB A}$ is divisible by 4, divisibility rules say that $\overline{BA}$ is divisible by 4, implying $B = 5$. Therefore,

$$A + B = 6 + 5 = \boxed{\text{(C) } 11}$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)